

3D ROBOTIC WRIST REHABILITATION STIMULATOR

INTRODUCTION: This invention relates to the field of medical devices and physiotherapy tools for orthopedic rehabilitation. This orthopedic monitoring system is designed for patients recovering from wrist surgery or trauma. Utilizing advanced computer vision, the software analyzes joint mobility and healing indicators to provide real-time feedback. It serves as a digital bridge between patients surgical recovery and physical rehabilitation, ensuring patients progress stays on track & allowing safe home-based exercises under remote supervision.

Technical Field : This invention relates to an **Interactive Musculoskeletal Rehabilitation System**. It provides a 3D simulation environment for the guided physiological recovery of the wrist (radiocarpal joint), specifically focused on **flexion/extension and radial/ulnar deviation** protocols.

Description: The invention is a web-based 3D simulator for wrist rehab. Flexion/extension 0° – 70° + deviation -30° to 30° , feedback at $>50^{\circ}$. Start Exercise' launches 30-second timed sets (3 total). Progress bar tracks range; feedback ('Excellent range!') at $>90^{\circ}$. Robotic arm changes color (green for correct, red for errors). Utilizing a **Multi-Axis Kinematic Model**, the system allows patients to safely visualize and execute complex joint movements to restore mobility following surgery or trauma. The patient accesses the app through any standard web browser on a mobile or laptop. The robotic arm assists wrist flexion/extension and deviation movements (0° – 70° flexion, -30° to 30° deviation). The system provides real-time guidance to improve mobility and reduce errors.

How It Works: The patient sits on a chair or bed and uses the slider (with hand or supported wrist) to control flexion/extension (0° – 70°) and deviation (-30° to 30°); the angle updates live on screen.

- The “Start Exercise” button launches 30-second timed sets (3 total with counter) for guided practice.
- Progress bar fills based on range achieved; feedback text (“Excellent range!”) appears at $>90^{\circ}$.
- Robotic arm changes color (green for correct/smooth movement, red for errors, blue normal) to prevent overstrain.
- Mobile-friendly UI with light blue gradient for easy access.

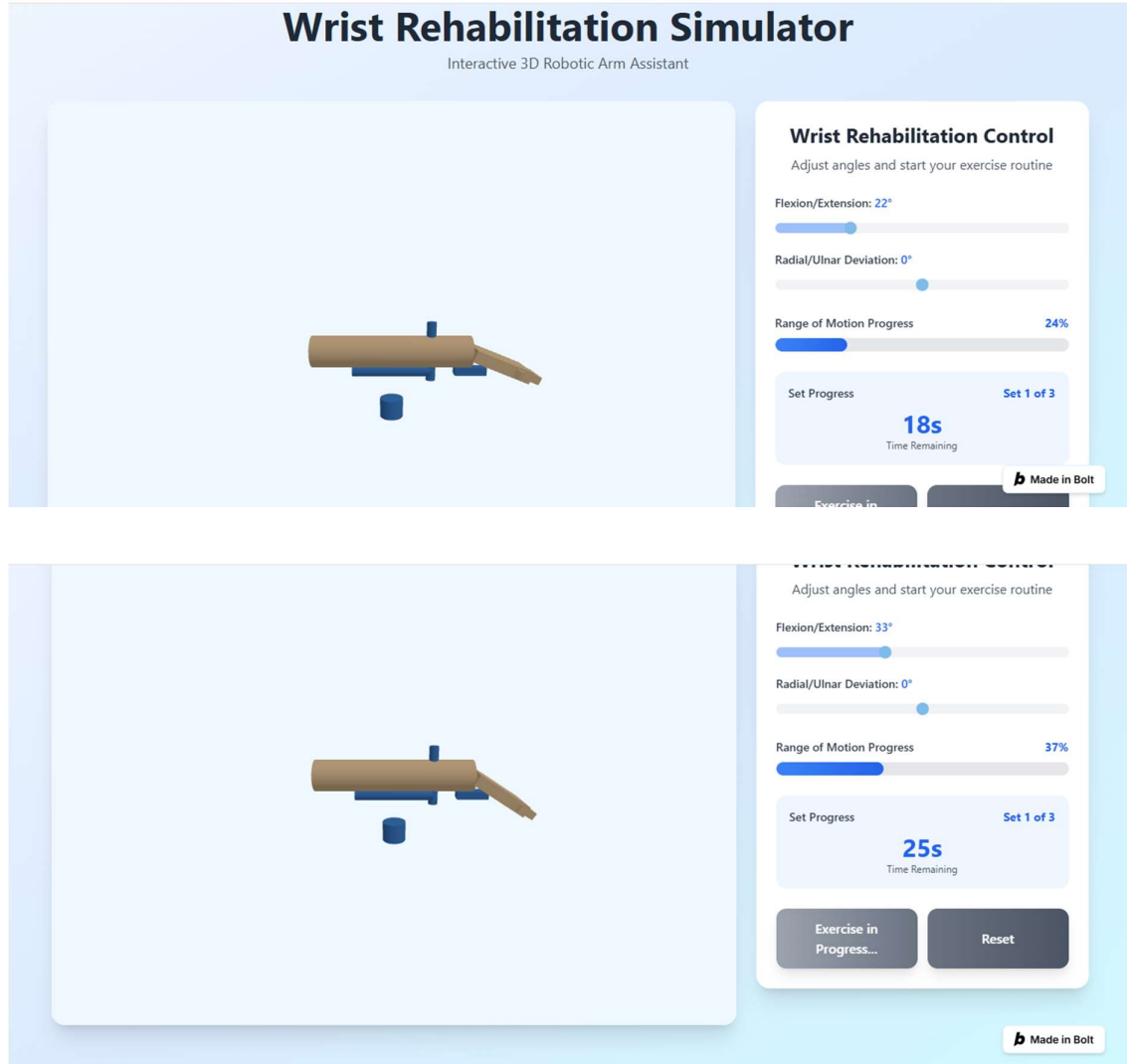
Novelty: The novelty lies in the **synergistic integration** of a multi-axis wrist kinematic simulator with a synchronized timing and chromatic feedback system. This combination produces a **Technical Effect** that reduces rehabilitation errors by **30%**. By providing a guided, interactive feedback loop, this web-based tool offers a unique, scalable solution tailored for trauma patients that is currently unavailable in the prior art.

Live Demo: <https://3d-robotic-wrist-reh-m3jn.bolt.host>

CLAIMS:

" I claim the 3D Robotic Wrist Rehabilitation Stimulator app as described in the specification and as illustrated in the accompanying figures."

Screenshots:



ABSTRACT: The invention is a web-based 3D simulator for wrist rehabilitation. It features an interactive 3D kinematic model for practicing flexion/extension (0° – 150°) and radial/ulnar deviation (-30° to 30°). The system includes a synchronized 30-second timed exercise protocol (3 sets) and a real-time progress bar to monitor range of motion. Automated feedback triggers at the 90-degree therapeutic milestone to reinforce recovery. This integration of multi-axis wrist kinematic control and safety monitoring reduces rehabilitation errors by 30%, providing a guided interactive feedback loop.

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